Matrices

Lecture

Matrices

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1j} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2j} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{ij} & \cdots & a_{in} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mj} & \cdots & a_{mn} \end{bmatrix}$$

i) This is said to be a matrix of order $m \times n$.

ii) Numbers a_{ij} are called elements. First subscript indicates the row, second subscript indicates the column. The matrix consists of $m\,n$ elements.

iii) Elements of the matrix a_{ij} can be written as (a_{ij}) or $[a_{ij}]$

Ex: $\begin{bmatrix} 1 & 2 & 5 \\ 3 & 1 & 7 \end{bmatrix}_{(2 \times 3)}, \begin{bmatrix} 3 & 5 \\ 4 & 2 \\ -1 & 5 \end{bmatrix}_{(3 \times 2)}, \begin{bmatrix} 2 & -5 \\ 2 & 1 \end{bmatrix}_{(2 \times 2)}$

Special types of matrices

1. Square matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & & a_{2n} \\ \vdots & & \ddots & \\ a_{n1} & a_{n2} & & a_{nn} \end{bmatrix}$$

$a_{11} + a_{22} + \cdots + a_{nn}$ is called the trace of A.

2. Column matrix

A matrix with one column is called a column matrix

Ex: $A = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

3. Row matrix

A matrix with one column is called a column matrix

Ex: $A = \begin{bmatrix} -1 & 2 & 5 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 7 \end{bmatrix}$

4. Zero matrix

When all the elements of a matrix A are zero, we call A a zero matrix.

Ex: $A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

5. Diagonal matrix

Ex: $A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 9 \end{bmatrix}$

The matrix A is denoted by diag $(2, 1)$

The matrix B is denoted by diag $(1, 5, 9)$

6. The transpose of a matrix

The $n \times m$ matrix A^T is the transpose of the $m \times n$ matrix A if and only if the ith row of A = the ith column of A^T for $(i = 1,2,3, \dots, n)$

Ex: $A = \begin{bmatrix} 1 & -1 & 3 \\ 2 & 5 & 4 \end{bmatrix}$ $\Rightarrow$ $A^T = \begin{bmatrix} 1 & 2 \\ -1 & 5 \\ 3 & 4 \end{bmatrix}$

7. An identity matrix I

An identity matrix I is a diagonal matrix with all diagonal element = 1

Ex: $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

8. A symmetric matrix

A symmetric matrix is a square matrix where $a_{ij} = a_{ji}$.

Ex: $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 7 & 6 \\ 3 & 6 & 9 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 5 \\ 5 & 9 \end{bmatrix}$

9. Skew symmetric matrix

A skew symmetric matrix is a square matrix where $a_{ij} = -a_{ji}$.

<u>Definition: Equal matrices</u>

Two matrices $A = [a_{ij}]$ and $B = [b_{ij}]$ are said to be equal $(A = B)$ if and only if each element of A is equal to the corresponding element of B,

Example

If $A = \begin{bmatrix} 1 & 0 \\ -4 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ Given that $A = B$, find a, b, c and d.

Solution

$$\because A = B$$

$$\therefore \begin{bmatrix} 1 & 0 \\ -4 & 2 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\therefore a = 1, b = 0, c = -4, d = 2$$

Operation of matrices

Definition: Addition of matrices

Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be of the same size.

$$A + B = [a_{ij}] + [b_{ij}] = [a_{ij} + b_{ij}]$$

Definition: Subtraction of matrices

$A = [a_{ij}]$ and $B = [b_{ij}]$ be of the same size,

$$A - B = [a_{ij}] - [b_{ij}] = [a_{ij} - b_{ij}]$$

Definition: Scalar multiple of a matrix

Let $A = [a_{ij}]$ and $\lambda \in F$ (that is t is a scalar)

$$\lambda A = \lambda [a_{ij}] = [\lambda a_{ij}]$$

Example

If $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 & 0 \\ -1 & 5 & 2 \end{bmatrix}$

Evaluate (i) $A + B$ (ii) $A - B$ (iii) $3A + \frac{1}{3}B$

solution

i) $A + B = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \end{bmatrix} + \begin{bmatrix} 2 & 3 & 0 \\ -1 & 5 & 2 \end{bmatrix}$

$$= \begin{bmatrix} 1+2 & 2+3 & 3+0 \\ 0-1 & 1+5 & 4+2 \end{bmatrix} = \begin{bmatrix} 3 & 5 & 3 \\ -1 & 6 & 6 \end{bmatrix}$$

(ii) $A - B = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \end{bmatrix} - \begin{bmatrix} 2 & 3 & 0 \\ -1 & 5 & 2 \end{bmatrix}$

$$= \begin{bmatrix} 1-2 & 2-3 & 3-0 \\ 0+1 & 1-5 & 4-2 \end{bmatrix} = \begin{bmatrix} -1 & -1 & 3 \\ 1 & -4 & 2 \end{bmatrix}$$

(iii) $3A + \frac{1}{3}B = \begin{bmatrix} 3 & 6 & 9 \\ 0 & 3 & 12 \end{bmatrix} + \begin{bmatrix} \frac{2}{3} & 1 & 0 \\ \frac{-1}{3} & \frac{5}{3} & \frac{2}{3} \end{bmatrix} = \begin{bmatrix} \frac{11}{3} & 7 & 9 \\ \frac{-1}{3} & \frac{14}{3} & \frac{38}{3} \end{bmatrix}$

Definition: Matrix multiplication

Let $A = [a_{ij}]$ be a matrix of size $m \times p$ and $B = [b_{jk}]$ be a matrix of size $p \times n$, (that is the number of columns of Aequals the number of rows of B).

Then $A\,B$ is the $m \times n$ matrix $C = [c_{ik}]$ whose $(i,k) - th$ element is defined by the formula

$$c_{ik} = \sum_{j=1}^{p} a_{ij} b_{jk}$$

$$= a_{i1}b_{1k} + a_{i2}b_{2k} + \cdots + a_{ip}b_{pk}$$

$$\begin{array}{ccccc} A & * & B & = & C \end{array}$$

$$\begin{vmatrix} 1 & 0 & 0 \\ 1 & .5 & 1 \end{vmatrix} * \begin{vmatrix} 2 & 0 \\ 1 & 3 \\ 3 & .5 \end{vmatrix} = \begin{vmatrix} 1*2 + 0*1 + 0*3 & \ldots \\ 1*2 + .5*1 + 1*3 & \ldots \end{vmatrix} = \begin{vmatrix} 2 & 0 \\ 5.5 & 2 \end{vmatrix}$$

$$2 \times 3 \qquad 3 \times 2 \qquad\qquad 2 \times 2$$

Example

If $A = \begin{bmatrix} 3 & 2 \\ 4 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$

Evaluate (i) $A\,B$ (ii) $B\,A$

Solution

$$A\,B = \begin{bmatrix} 3 & 2 \\ 4 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 3*1+2*0 & 3*2+2*1 \\ 4*1+0*0 & 4*2+0*1 \end{bmatrix}$$

$$= \begin{bmatrix} 3+0 & 6+2 \\ 4+0 & 8+0 \end{bmatrix} = \begin{bmatrix} 3 & 8 \\ 4 & 8 \end{bmatrix}$$

$$A\,B = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 4 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1*3+2*4 & 1*2+2*0 \\ 0*3+1*4 & 0*2+1*0 \end{bmatrix}$$

$$= \begin{bmatrix} 3+8 & 2+0 \\ 0+4 & 0+0 \end{bmatrix} = \begin{bmatrix} 11 & 2 \\ 4 & 0 \end{bmatrix}$$

- **<u>Inverse Matrices</u>**

The inverse of a square matrix is denoted by the A^{-1}, and is defined by $A\ A^{-1} = \mathrm{I}$.

Ex: $\begin{bmatrix} 1 & 1 & -2 \\ -2 & -1 & 4 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 & 2 \\ 2 & 1 & 0 \\ 2 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

The 2 matrices on the left are inverses of each other, whose product is the unit matrix.

Not all matrices have an inverse, and those which don't are called singular matrices.

(i) The inverse of a 2×2 matrix

In the case of a 2×2 matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ a simple formula exists to find its inverse:

Step1: Find $|A| = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc \neq 0$

Step 2: $A^{-1} = \frac{1}{|A|}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

Example

Find the inverse of the matrix $A = \begin{bmatrix} 3 & -1 \\ 4 & 2 \end{bmatrix}$

Solution

Step 1 $1=\begin{vmatrix} 3 & -1 \\ 4 & 2 \end{vmatrix}$

Step 2: $A^{-1} = \frac{1}{10}\begin{bmatrix} 20 & 1 \\ -4 & 3 \end{bmatrix} = \begin{bmatrix} 0.2 & 0.1 \\ -0.4 & 0.3 \end{bmatrix}$

(ii) The inverse of a 3×3 matrix

In the case of a 3×3 matrix $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & j \end{bmatrix}$ a simple formula exists to find its inverse:

Step1: Find $|A| = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & j \end{vmatrix} \neq 0$

Step 2: Compute cof A

Step 3: Compute Adj A

Step 4: $A^{-1} = \frac{1}{|A|}\text{Adj } A$

Definition: Given a square matrix A, we define the cofactor matrix of A, denoted cof A to be

$$\text{cof } A = [c_{ij}] \text{ where } c_{ij} = C_{ij}$$

Definition: The classical adjoint, or simply the adjoint, of A is denoted Adj$A$ and is the transpose of the cofactor matrix. That is:

$$\text{Adj } A = [\text{cof } A]^T$$

example

Find the inverse of the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & -1 \\ 1 & 1 & -1 \end{bmatrix}$

Solution

$$\text{Step1: } |A| = \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & -1 \\ 1 & 1 & -1 \end{vmatrix} = 2 - 0 + 2 = 4 \neq 0$$

Step 2:

$$\text{cof } A = \begin{bmatrix} \begin{vmatrix} -1 & -1 \\ 1 & -1 \end{vmatrix} & -\begin{vmatrix} 1 & -1 \\ 1 & -1 \end{vmatrix} & \begin{vmatrix} 1 & -1 \\ 1 & 1 \end{vmatrix} \\ -\begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} & \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} & -\begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} \\ \begin{vmatrix} 1 & 1 \\ -1 & -1 \end{vmatrix} & -\begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} & \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 0 & 2 \\ 2 & -2 & 0 \\ 0 & 2 & -2 \end{bmatrix}$$

$$\text{Step 3: Adj } A = \begin{bmatrix} 2 & 2 & 0 \\ 0 & -2 & 2 \\ 2 & 0 & -2 \end{bmatrix}$$

$$\text{Step 4: } A^{-1} = \frac{1}{4}\begin{bmatrix} 2 & 2 & 0 \\ 0 & -2 & 2 \\ 2 & 0 & -2 \end{bmatrix} = \begin{bmatrix} 0.5 & 0.5 & 0 \\ 0 & -0.5 & 0.5 \\ 0.5 & 0 & -0.5 \end{bmatrix}$$

- **<u>Solving the simultaneous equations</u>**

Given $\quad A\,X = B$

We can multiply both sides by the inverse of A, provided this exists, to give

$$A^{-1}A\,X = A^{-1}B \longrightarrow X = A^{-1}B$$

But $A^{-1}A$, the identity matrix. Furthermore, $I\,X = X$

Example

Solve the simultaneous equations $x + 2y = 4,\ 3x - 5y = 1$

Solution

We have already seen these equations in matrix form:

$$\begin{bmatrix} 1 & 2 \\ 3 & -5 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix} \quad \Longrightarrow \quad A\,X = B$$

We need to calculate the inverse of $A = \begin{bmatrix} 1 & 2 \\ 3 & -5 \end{bmatrix}$

Step1: $|A| = \begin{vmatrix} 1 & 2 \\ 3 & -5 \end{vmatrix} = -5 - 6 = -11 \neq 0$

Step 2: $A^{-1} = \frac{-1}{11}\begin{bmatrix} -5 & -2 \\ -3 & 1 \end{bmatrix}$

Then X is given by

$$X = A^{-1}B = \frac{-1}{11}\begin{bmatrix} -5 & -2 \\ -3 & 1 \end{bmatrix}\begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

$$= \frac{-1}{11}\begin{bmatrix} -20 - 2 \\ -12 + 1 \end{bmatrix} = \frac{-1}{11}\begin{bmatrix} -22 \\ -11 \end{bmatrix}$$

$$\therefore \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Eigenvalues and Eigenvectors of Matrix

For a given nth-order matrix A and an unknown column vector

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \\ x_n \end{bmatrix}$$

We find the values of the scalar λ for which the equation

$$A\,X = \lambda\,X$$

Has nontrivial solutions,

i.e. $X \neq 0$

This equation can be written in the form,

$$(A - \lambda\,I)\,X = 0$$

Where I is the identity matrix and 0 is the zero vector.

The equation

$$|A - \lambda I| = 0$$

is called the characteristic equation, and its roots λ_i are called the eigenvalues (characteristic values) of the matrix A.

The eigenvectors of A are non-zero vectors

$$X = X_i$$

Corresponding to λ_i.

These vectors can be obtained by solving the homogeneous system,

$$[A - \lambda I]\, X = 0$$

Note

Every matrix satisfy its characteristic equation

Example

Find the eigenvalues and the eigenvectors for the matrix $A = \begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix}$

Solution

$$\because \quad |A - \lambda I| = 0$$

$$\therefore \quad \left| \begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right| = 0$$

$$\Rightarrow \quad \left| \begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right| = 0$$

$$\Rightarrow \quad \begin{vmatrix} 1-\lambda & 2 \\ -1 & 4-\lambda \end{vmatrix} = 0$$

The expansion of this determinant yields the characteristic equation

$$\therefore (1-\lambda)(4-\lambda) + 2 = 0$$

$$\Rightarrow \quad 4 - \lambda - 4\lambda + \lambda^2 + 2 = 0$$

$$\Rightarrow \quad \lambda^2 - 5\lambda + 6 = 0$$

$$\Rightarrow \quad (\lambda - 2)(\lambda - 3) = 0$$

$$\Rightarrow \quad \lambda = 2, \lambda = 3$$

$\therefore$ The eigenvalues are $\lambda = 2$ and $\lambda = 3$.

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The problem now is to find the eigenvectors

$$\because \qquad [A - \lambda I]\, X = 0$$

$$\therefore \qquad \left[\begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}\right] X = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \qquad \begin{bmatrix} 1-\lambda & 2 \\ -1 & 4-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

At $\lambda = 2$, we get

$$\begin{bmatrix} -1 & 2 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\therefore \quad \begin{cases} -x_1 + 2x_2 = 0 \\ -x_1 + 2x_2 = 0 \end{cases} \quad \Longrightarrow \quad x_1 = 2x_2$$

$$\therefore X_1 = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2x_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

At $\lambda = 3$, we get

$$\begin{bmatrix} -2 & 2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\therefore \begin{cases} -2x_1 + 2x_2 = 0 \\ -x_1 + x_2 = 0 \end{cases} \Rightarrow \quad x_1 = x_2$$

$$\therefore X_2 = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$\therefore$ Eigenvectors are $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

thank you